Chapter 9F

Neurology Review: Integrated Cases, Diagnostic Algorithms, and Graph-RAG Map

Medvale Internal Medicine - Neurology for Internal Medicine

This final neurology chapter is a synthesis section. It ties together localization, stroke/TIA/hemorrhage, altered mental status, syncope, seizures, movement disorders, dementia, demyelinating disease, neuromuscular disease, spinal cord syndromes, vertigo, and neurocutaneous disorders. It is designed to be case-heavy and extraction-friendly for Medvale Graph-RAG.

1. The master neurologic reasoning algorithm

Neurology becomes manageable when every presentation is forced through the same sequence: time course, localization, syndrome category, dangerous mimics, diagnostic confirmation, treatment, and follow-up prevention. Time course separates vascular events from inflammatory, infectious, toxic-metabolic, degenerative, and functional syndromes. Localization separates cortex, brainstem, spinal cord, peripheral nerve, neuromuscular junction, muscle, vestibular system, and systemic encephalopathy.

Master first-pass algorithm

Step	Question	High-yield implication
1. Tempo	Seconds-minutes, hours-days, weeks-months, or years?	Sudden focal deficits are vascular until proven otherwise; relapsing deficits suggest demyelination; chronic decline suggests neurodegeneration.
2. Consciousness	Is awareness or attention impaired?	AMS/coma pushes toward metabolic, toxic, infection, seizure, hypoxia, or diffuse structural disease.
3. Focality	Is there a focal deficit?	Focal deficit localizes to CNS or nerve; nonfocal confusion often reflects systemic brain dysfunction.
4. Level	Cortex, brainstem, cord, root, nerve, NMJ, muscle, vestibular?	Aphasia/neglect = cortex; crossed signs = brainstem; sensory level/bladder = cord; fatigability = NMJ.
5. Emergency screen	Could this kill or permanently disable today?	Stroke, ICH, SAH, meningitis, status epilepticus, GBS respiratory failure, myasthenic crisis, cord compression.
6. Confirmation	What test answers the next branch question?	CT for hemorrhage, CTA for LVO, MRI for cord/MS, LP for CNS infection/SAH/MS, EEG for seizure/status, EMG/NCS for peripheral/NMJ.

2. Localization cheat sheet

Localization by bedside clue

Finding	Most likely localization	Reasoning
Aphasia	Dominant cortex, usually MCA territory	Language disturbance localizes above the brainstem and is usually cortical.
Neglect or anosognosia	Nondominant parietal/frontal cortex	Attention to one side of space is a cortical network function.
Diplopia + dysarthria + ataxia	Brainstem/cerebellar posterior circulation	Cranial nerve plus long tract or cerebellar findings.
Pure motor hemiparesis	Internal capsule, corona radiata, pons	Lacunar pathway if no cortical signs.
Sensory level + bladder dysfunction	Spinal cord	A horizontal level and autonomic dysfunction are cord signs.
Ascending weakness + areflexia	Peripheral nerve roots	GBS pattern until proven otherwise.
Fatigable ptosis/diplopia	Neuromuscular junction	Fluctuation with preserved sensation/reflexes suggests MG.
Proximal weakness + high CK	Muscle	Myopathic pattern; evaluate inflammatory, toxic, endocrine, dystrophic causes.
Vertigo + vertical nystagmus	Central vestibular/brainstem/cerebellar	Vertical or direction-changing nystagmus is central until proven otherwise.

Finding	Most likely localization	Reasoning
Fluctuating attention	Diffuse brain dysfunction	Delirium rather than isolated focal lesion.

3. Emergency neurologic syndromes

The fastest way to review neurologic emergencies is to ask what will be missed if the clinician waits. Stroke and LVO lose penumbra. ICH expands. SAH rebleeds. Status epilepticus injures neurons and causes systemic collapse. GBS and myasthenic crisis cause respiratory failure. Cord compression causes permanent paralysis and sphincter dysfunction. Meningitis and encephalitis kill by inflammation, edema, sepsis, and seizures.

Do-not-miss neurologic emergencies

Emergency	Key clue	Immediate action
Acute ischemic stroke / LVO	Sudden focal deficit, aphasia, neglect, hemiparesis, brainstem syndrome.	Noncontrast CT, CTA when LVO possible, reperfusion pathway.
Intracerebral hemorrhage	Focal deficit with headache, vomiting, depressed consciousness, severe HTN or anticoagulation.	CT, ICU, BP control, anticoagulant reversal, neurosurgical triage.
Subarachnoid hemorrhage	Thunderclap worst headache, meningismus, syncope, vomiting.	CT; if high suspicion persists, LP/CTA pathway; secure aneurysm; nimodipine.
Status epilepticus	Seizure >5 minutes or recurrent seizures without baseline recovery.	ABCs/glucose, benzodiazepine, antiseizure loading, ICU/EEG if persistent.
Meningitis/encephalitis	Fever, headache, neck stiffness, AMS, seizure, immunosuppression.	Blood cultures, empiric antimicrobials, LP when safe.
GBS respiratory failure	Ascending weakness, areflexia, weak cough, bulbar symptoms.	Hospitalize, monitor FVC/NIF, IVIG or plasma exchange.
Myasthenic crisis	MG with dyspnea, dysphagia, weak cough, low FVC.	ICU, respiratory support, IVIG/PLEX, treat trigger.
Cord compression/myelitis	Back pain, weakness, sensory level, urinary retention.	Urgent MRI spine and cause-specific treatment.

4. Stroke and hemorrhage rapid review

Stroke decision table

Presentation	Likely syndrome	Most useful next branch
Aphasia and right arm/face weakness	Dominant MCA ischemic stroke until proven otherwise.	CT to exclude bleed, CTA for LVO, reperfusion eligibility.
Pure motor weakness without aphasia/neglect	Lacunar infarct possible.	Still acute stroke pathway; later focus on HTN/DM control.
Transient monocular blindness	Carotid-territory TIA.	Urgent vascular imaging and secondary prevention.
Focal deficit + hyperdense deep bleed	Hypertensive ICH likely.	ICU, BP control, reversal if anticoagulated, ICP monitoring.
Worst headache of life + neck stiffness	SAH until proven otherwise.	CT then LP/CTA pathway if needed; aneurysm treatment/nimodipine.
Vertigo, dysarthria, inability to walk	Posterior circulation stroke possible.	Do not label benign vertigo; image posterior circulation.

- Ischemic stroke: the acute branch is hemorrhage excluded, time known, disabling deficit present, LVO present or absent.
- TIA: resolved symptoms still require urgent evaluation because early recurrence risk is front-loaded.
- ICH: the acute branch is hematoma size/location, intraventricular extension, anticoagulation, BP, hydrocephalus, neurosurgical need.
- SAH: thunderclap headache demands follow-through even if initial CT is negative after a delayed presentation.

5. AMS, syncope, and seizure rapid review

Transient or altered consciousness comparison

Syndrome	Core mechanism	Clues
Delirium	Diffuse brain dysfunction from medical illness, drugs, infection, hypoxia, metabolic disorder.	Acute fluctuating attention failure; visual hallucinations; worse at night.
Coma	Severe arousal failure from structural or toxic-metabolic cause.	Pupils, brainstem reflexes, motor response, respiratory pattern guide localization.
Syncope	Transient cerebral hypoperfusion.	Prodrome and rapid recovery; exertional/supine/no prodrome suggests cardiac danger.
Seizure	Abnormal cortical discharge.	Aura, tonic-clonic activity, lateral tongue bite, postictal confusion; provoked vs unprovoked matters.
Nonconvulsive status	Ongoing seizure activity without convulsions.	Persistent unexplained AMS after convulsions or in high-risk patient; EEG diagnosis.

Exam trap: brief jerking can occur in syncope. Do not diagnose epilepsy solely from a few myoclonic movements after fainting; ask about prodrome, duration, recovery, tongue bite location, triggers, and witness description.

6. Movement disorders and dementia rapid review

Movement disorder pattern table

Pattern	Likely diagnosis	Distinguishing point
Rest tremor + bradykinesia + rigidity	Parkinson disease	Bradykinesia is required; levodopa is most effective symptomatic therapy.
Parkinsonism + early falls + vertical gaze palsy	Progressive supranuclear palsy	Atypical parkinsonism; often poor levodopa response.
Parkinsonism + severe autonomic failure	Multiple system atrophy	Orthostasis, urinary dysfunction, erectile dysfunction, cerebellar signs.
Action/postural tremor, alcohol-responsive	Essential tremor	No rigidity or bradykinesia.
Chorea + psychiatric change + dementia	Huntington disease	Autosomal dominant CAG repeat expansion.
Ataxia + nystagmus + dysmetria	Cerebellar syndrome	Evaluate stroke, tumor, alcohol, MS, vitamin deficiency, inherited disease.

Dementia pattern table

Pattern	Likely diagnosis	Distinguishing point
Insidious memory-predominant decline	Alzheimer disease	Most common dementia; age is major risk factor.
Stepwise decline + focal deficits	Vascular dementia	Stroke/small-vessel burden; vascular risk control matters.
Visual hallucinations + fluctuations + parkinsonism	Dementia with Lewy bodies	Neuroleptic sensitivity.
Personality/language change before memory loss	Frontotemporal dementia	Disinhibition, apathy, compulsions, aphasia variants.
Gait impairment + urinary symptoms + cognitive decline	Normal pressure hydrocephalus	Potentially reversible; ventriculomegaly.
Low mood + poor concentration	Depression/pseudodementia	Treat mood disorder; can coexist with dementia.

7. Demyelinating, neuromuscular, and spinal cord rapid review

Weakness and sensory syndrome table

Clinical script	Likely localization	Diagnosis to consider
Relapsing optic neuritis, INO, sensory episodes	CNS white matter	MS, NMOSD, MOGAD, mimics.
Ascending weakness + areflexia after infection	Peripheral roots/nerves	Guillain-Barre syndrome.
Fatigable ptosis, diplopia, dysphagia	Neuromuscular junction	Myasthenia gravis.
Proximal leg weakness that improves with exertion + dry mouth	Presynaptic NMJ	Lambert-Eaton syndrome; search for small cell lung cancer.

Clinical script	Likely localization	Diagnosis to consider
Sensory level + urinary retention	Spinal cord	Transverse myelitis, cord compression, infarct.
Cape-like pain/temp loss	Central cervical cord	Syringomyelia.
Ipsilateral weakness/proprioception loss + contralateral pain/temp loss	Hemisection cord	Brown-Sequard syndrome.

8. Vertigo and gait algorithm

Dizziness is a symptom, not a diagnosis. The important distinction is whether the patient has benign peripheral vertigo or central vestibular disease. Brief positional spells suggest BPPV. Acute continuous vertigo with nausea and no hearing loss may be vestibular neuritis, but central causes must be considered when there are neurologic deficits or severe gait disturbance. Hearing loss and tinnitus suggest peripheral labyrinth/cochlear involvement, but AICA stroke can rarely produce hearing symptoms.

Vertigo review

Feature	Peripheral	Central
Nystagmus	Unidirectional horizontal-torsional, suppresses with fixation.	Vertical, direction-changing, pure torsional, not suppressed.
Neurologic signs	Absent except hearing symptoms in selected syndromes.	Diplopia, dysarthria, dysphagia, ataxia, weakness, numbness.
Gait	Unsteady but often able to walk.	Severe truncal ataxia or inability to walk.
Common examples	BPPV, vestibular neuritis, Meniere disease.	Cerebellar stroke, brainstem stroke, MS, tumor.
Best next step if dangerous	ENT/vestibular treatment if classic benign pattern.	Stroke pathway or MRI/vascular imaging.

9. Integrated cases

Case 1: Aphasia after breakfast

A 73-year-old suddenly cannot speak and has right face and arm weakness. Glucose is normal.

Reasoning: Dominant MCA stroke until proven otherwise. Noncontrast CT, CTA for LVO, reperfusion pathway.

Case 2: Worst headache

A 44-year-old has a thunderclap headache, vomiting, photophobia, and neck stiffness.

Reasoning: SAH is the dangerous diagnosis. CT first; if negative but suspicion remains, LP/CTA pathway.

Case 3: Quiet older adult

An 82-year-old becomes inattentive and withdrawn after starting diphenhydramine for sleep.

Reasoning: Hypoactive delirium from anticholinergic exposure is likely; remove trigger and search for other causes.

Case 4: Exertional collapse

A 39-year-old faints while running, with no prodrome.

Reasoning: High-risk cardiac syncope; ECG and structural/rhythm evaluation are needed.

Case 5: Convulsions stop but coma persists

A patient stops shaking after benzodiazepine but remains unresponsive for a prolonged period.

Reasoning: Consider nonconvulsive status epilepticus; EEG and continued evaluation are needed.

Case 6: Rest tremor

A 68-year-old has asymmetric pill-rolling rest tremor, bradykinesia, micrographia, and shuffling gait.

Reasoning: Parkinson disease is likely; review dopamine-blocking meds and consider levodopa if functionally impaired.

Case 7: Visual hallucinations

A 76-year-old has fluctuating cognition, visual hallucinations, and mild parkinsonism. Haloperidol worsens rigidity.

Reasoning: Dementia with Lewy bodies with neuroleptic sensitivity.

Case 8: Ascending weakness

A 32-year-old develops ascending weakness and areflexia after diarrhea; he has a weak cough.

Reasoning: GBS with respiratory risk. Admit, monitor FVC/NIF, treat with IVIG or plasma exchange.

Case 9: Fatigable diplopia

A 45-year-old has ptosis and diplopia worse at the end of the day, plus dysphagia.

Reasoning: Myasthenia gravis. Check antibodies/EMG, image thymus, monitor respiratory mechanics if bulbar symptoms.

Case 10: Sensory level

A patient develops bilateral leg weakness, numbness below the umbilicus, and urinary retention.

Reasoning: Spinal cord syndrome. Urgent MRI spine for transverse myelitis, compression, infarct, abscess, or tumor.

10. Common traps that lose points or harm patients

- Do not call a TIA benign because symptoms resolved.
- Do not lower BP aggressively in ischemic stroke unless thrombolysis protocol or another emergency requires it.
- Do not give antithrombotics before excluding hemorrhage.
- Do not ignore posterior circulation stroke in a patient with vertigo plus dysarthria, diplopia, ataxia, or inability to walk.
- Do not assume all shaking is seizure; syncope can have brief myoclonic jerks.
- Do not diagnose dementia during an acute delirium without reassessing after the medical trigger improves.
- Do not use dopamine blockers casually in Lewy body dementia or Parkinson disease.
- Do not wait for hypoxia in GBS or MG; monitor ventilatory mechanics.
- Do not diagnose MS from vague symptoms alone; require objective CNS demyelination pattern and exclusion of mimics.
- Do not miss spinal cord disease when there is a sensory level, sphincter dysfunction, or bilateral long-tract signs.

11. Graph-RAG integrated extraction map

Core neurology nodes

Node	Class	Key relationships
Acute ischemic stroke	Vascular disease	presents_with focal deficit; diagnosed_by CT/CTA/MRI; treated_by thrombolysis/thrombectomy/secondary prevention.
Intracerebral hemorrhage	Vascular disease	presents_with focal deficit/headache/vomiting; treated_by BP control/reversal/ICP/neurosurgery.
Subarachnoid hemorrhage	Vascular disease	presents_with thunderclap headache; diagnosed_by CT/LP/CTA; treated_by aneurysm securing/nimodipine.
Delirium	Cognitive syndrome	caused_by infection/drugs/metabolic/hypoxia; distinguished_from dementia by acute fluctuation and inattention.
Syncope	Transient LOC syndrome	caused_by cerebral hypoperfusion; classified_as reflex/orthostatic/cardiac/PE/rare cerebrovascular.
Seizure	Paroxysmal neurologic event	classified_as focal/generalized; caused_by metabolic/structural/infectious/toxic; diagnosed_by history/EEG.
Parkinson disease	Movement disorder	presents_with bradykinesia/rest tremor/rigidity; treated_by levodopa and other dopaminergic strategies.
Dementia	Cognitive syndrome	subtyped_as Alzheimer/vascular/DLB/FTD/NPH; distinguished_from delirium and depression.
Multiple sclerosis	Demyelinating disease	presents_with optic neuritis/INO/relapsing deficits; diagnosed_by MRI/CSF; treated_by steroids for relapse and DMTs.
Guillain-Barre syndrome	Peripheral nerve disease	presents_with ascending weakness/areflexia; complicated_by respiratory failure; treated_by IVIG/PLEX.
Myasthenia gravis	NMJ disease	presents_with fatigable weakness; diagnosed_by antibodies/EMG; treated_by pyridostigmine/immunotherapy/thymectomy/IVIG/PLEX.

Node	Class	Key relationships
Spinal cord syndrome	Localization class	presents_with sensory level/bilateral weakness/sphincter dysfunction; diagnosed_by MRI spine.
Central vertigo	Vestibular localization	suggested_by vertical nystagmus/focal deficits/severe ataxia; evaluated_as posterior circulation emergency.

12. Final practice set

Q1. Aphasia localizes to what broad region?

Answer: Dominant cortex, commonly MCA territory.

Q2. What diagnosis is suggested by thunderclap headache and xanthochromia?

Answer: Subarachnoid hemorrhage.

Q3. What is the defining cognitive feature of delirium?

Answer: Acute fluctuating inattention/awareness disturbance.

Q4. Which syncope features are most concerning?

Answer: Exertional, supine, no prodrome, palpitations, abnormal ECG, structural heart disease.

Q5. What is the first-line emergency drug class for active convulsive status epilepticus?

Answer: Benzodiazepines after ABCs/glucose.

Q6. Which movement feature is required for Parkinson disease diagnosis in most frameworks?

Answer: Bradykinesia.

Q7. Which dementia has visual hallucinations and neuroleptic sensitivity?

Answer: Dementia with Lewy bodies.

Q8. Which disorder causes ascending weakness and areflexia after infection?

Answer: Guillain-Barre syndrome.

Q9. What neuromuscular disease causes fatigable ptosis and diplopia?

Answer: Myasthenia gravis.

Q10. What finding makes vertigo central until proven otherwise?

Answer: Vertical or direction-changing nystagmus, focal neurologic signs, or severe truncal ataxia.

13. Source and clinical cross-check note

This capstone chapter was generated as original synthesis from the uploaded neurology source sections covering stroke/TIA/ICH/SAH, movement disorders, dementia, AMS/coma, syncope, seizures, demyelinating disease, neuromuscular disease, neurocutaneous syndromes, spinal cord disorders, and vertigo. It is intended for medical education and Graph-RAG indexing, not as a replacement for local emergency, neurology, neurosurgery, or critical care protocols.